

Week 1: Introduction & String Manipulation

Models of Language and Conversation

PSYC 51.07: Models of Language and Communication

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Today's Journey

- 1 Course Introduction
- 2 Is ChatGPT Conscious?
- 3 Language vs. Thought
- 4 Pattern Matching & String Manipulation
- 5 ELIZA: The First Chatbot
- 6 Assignment 1
- 7 Wrap-up

Welcome!

What is this course about?

We'll explore how machines can understand and generate human language:

- Building conversational agents from scratch
- Understanding how language relates to thought
- Hands-on programming with real models
- Critical thinking about AI consciousness and capabilities
- Research applications in cognitive science

This course is experiential!

You'll learn by **doing**: coding, experimenting, discussing, and researching.

Course Structure

What we'll build:

- 1 ELIZA (pattern matching)
- 2 SPAM classifier
- 3 Wikipedia analyzer
- 4 Customer service chatbot
- 5 GPT model

Grading:

- 5 Problem Sets: 75%
- Final Project: 25%

Tools we'll use:

- Google Colaboratory
- HuggingFace Models
- GitHub
- Discord

The Big Questions

Throughout this course, we'll grapple with:

- 1 Can machines truly *understand* language?
- 2 What is the relationship between language and thought?
- 3 How do statistical patterns capture meaning?
- 4 Can AI be conscious? Should we care?
- 5 What can language models tell us about human cognition?

Think about it!

These aren't just philosophical questions—they're at the heart of cognitive science and AI research!

Discussion: Is ChatGPT Conscious?

Discussion

Let's start with a provocative question:

Is ChatGPT conscious?

Think about:

- What does "conscious" even mean?
- How would we test for consciousness?
- Does it matter if ChatGPT *seems* conscious?
- What evidence would convince you either way?

What is Consciousness?

Some perspectives:

- **Phenomenal consciousness:** Subjective experience ("what it's like")
- **Access consciousness:** Information available for reasoning
- **Self-awareness:** Knowledge of one's own mental states
- **Sentience:** Ability to feel/perceive

Think about it!

If ChatGPT says "I feel happy," does it actually *feel* anything?
How is this different from a parrot saying "I'm hungry"?

The Hard Problem

Why is this so difficult?

We can't directly observe consciousness—even in other humans!

With humans, we assume consciousness because:

- We share similar biology
- We have our own conscious experience
- They behave in ways consistent with having experiences

But with AI:

- Completely different "biology" (silicon vs. neurons)
- No shared evolutionary history
- Can produce human-like behavior
- Does the substrate matter?

Current Scientific Consensus

Important Note

Most cognitive scientists and AI researchers agree: **Current LLMs are not conscious.**

Why not?

- No sensory-motor grounding in the world
- No persistent self-model or goals
- No phenomenal experience (as far as we can tell)
- Pattern matching $\neq$ understanding

Think about it!

But this raises another question: Is consciousness *necessary* for language? Or can you have sophisticated language without consciousness?

Language and Thought

Discussion

Do you need language to think?
Does language *shape* how you think?

Two extreme positions:

- ① **Language = Thought:** All thinking happens in language
 - Implies: No language, no complex thought
- ② **Language $\neq$ Thought:** Language is just a tool for communication
 - Implies: Rich mental life exists independently of language

The truth is likely somewhere in between...

Evidence: The Language Network

Fedorenko et al. (2024) - Nature

The language network as a natural kind within the broader landscape of the human brain

Key findings:

- The brain has a **specialized language network**
- This network is distinct from:
 - General reasoning/problem-solving
 - Mathematical thinking
 - Social cognition
 - Musical processing
- Language network activates for linguistic tasks, but not for non-linguistic thinking

Implication: Language and thought are *separable* in the brain!

Evidence: Language Shapes Perception

Lupyan et al. (2020) - Trends in Cognitive Sciences

Effects of language on visual perception

Key findings:

- Having words for things affects how we see them
- Language can:
 - Speed up visual search
 - Alter color perception
 - Influence object categorization
 - Modulate attention
- Example: Russian speakers (with different words for light/dark blue) perceive blue shades more distinctly

Implication: Language doesn't just describe thought—it *shapes* it!

So What About LLMs?

Think about it!

If language and thought are separable...

Can an LLM have sophisticated language *without* sophisticated thought?

Can it have sophisticated thought *without* language-like representations?

Current evidence suggests:

- LLMs are *very good* at language patterns
- LLMs may have some internal "representations" of concepts
- LLMs lack grounding in sensory-motor experience
- LLMs don't have persistent goals or self-models
- No evidence of phenomenal consciousness

How Do Computers Process Language?

The Fundamental Challenge

Computers don't "understand" meaning—they manipulate **strings of characters**.

Everything starts with patterns:

- 1 Finding specific words or phrases
- 2 Replacing parts of text
- 3 Extracting information
- 4 Transforming input to output

Even sophisticated models like GPT are (at their core) doing *pattern matching*—just at a much more complex level!

Basic String Manipulation

Example 1: Simple replacement

```
user_input = "I am feeling sad"
response = user_input.replace("I am", "Why are you")
print(response)
# Output: "Why are you feeling sad"
```

Example 2: Pattern matching with wildcards

```
import re

pattern = r"I am (.*)"
match = re.search(pattern, "I am feeling sad")
if match:
    feeling = match.group(1) # Captures "feeling sad"
    response = f"Why are you {feeling}?"
    print(response)
# Output: "Why are you feeling sad?"
```

More Complex Patterns

Capturing multiple parts:

```
pattern = r"I (.*) my (.*)"
match = re.search(pattern, "I love my family")

if match:
    verb = match.group(1)      # "love"
    object = match.group(2)    # "family"
    response = f"Tell me more about your {object}."
    print(response)
# Output: "Tell me more about your family."
```

Think about it!

This is remarkably simple, yet it can create the *illusion* of understanding!

Substitutions: Pre and Post

Pre-substitutions: Normalize input before matching

```
pre_subs = {"dont": "don't", "cant": "can't",  
            "wont": "won't"}  
  
user_input = "I dont know"  
for old, new in pre_subs.items():  
    user_input = user_input.replace(old, new)  
# Result: "I don't know"
```

Post-substitutions: Fix grammar in responses

```
post_subs = {"i": "you", "my": "your",  
             "am": "are", "I": "you"}  
  
response = "Why are you sad" # Already processed  
for old, new in post_subs.items():  
    response = response.replace(old, new)
```

The Power of Simple Patterns

Surprising Fact

With just pattern matching and string manipulation, you can create a chatbot that seems remarkably "intelligent"!

Why does this work?

- Humans are pattern-seeking creatures
- We *project* understanding onto machines
- Context and expectations shape interpretation
- Language has inherent structure and regularities

Important Lesson

"Intelligence" is often in the eye of the beholder. Just because something *seems* intelligent doesn't mean it truly understands!

ELIZA: A Revolutionary Program

Weizenbaum (1966)

ELIZA — A Computer Program For the Study of Natural Language Communication Between Man and Machine

What was ELIZA?

- First "chatbot" (1964-1966)
- Simulated a Rogerian psychotherapist
- Used pattern matching & string manipulation
- Created by Joseph Weizenbaum at MIT

Think about it!

ELIZA was meant to *demonstrate* the superficiality of human-computer interaction. Instead, people became emotionally attached to it!

How ELIZA Worked

The basic algorithm:

- 1 **Read** user input
- 2 **Pre-process**: Apply pre-substitutions
- 3 **Match**: Find highest-priority pattern
- 4 **Decompose**: Extract parts of input
- 5 **Reassemble**: Generate response using template
- 6 **Post-process**: Apply post-substitutions
- 7 **Output** response

No Understanding Required!

ELIZA had **zero** understanding of what it was saying. It was pure pattern matching!

Example ELIZA Conversation

Sample Interaction

ELIZA: Welcome. What brings you here today?

User: I am feeling sad.

ELIZA: Why do you feel sad?

User: I can't stop thinking about my family.

ELIZA: Tell me more about your family.

User: My mother never understood me.

ELIZA: Who else in your family never understood you?

User: I have to go now.

ELIZA: That will be \$200. See you again next week.

The ELIZA Effect

Definition: ELIZA Effect

The tendency of humans to attribute human-like understanding to computer programs, even when they know the program is simple.

What happened:

- People formed emotional bonds with ELIZA
- Users shared intimate details of their lives
- Some refused to believe it was "just" a program
- Weizenbaum's secretary asked him to leave the room so she could talk to ELIZA privately!

Think about it!

Why are we so eager to see intelligence and understanding in machines?

Weizenbaum's Warning

A Cautionary Tale

Joseph Weizenbaum became **deeply concerned** about people's reactions to ELIZA.

His concerns:

- People mistook simulation for understanding
- Risk of replacing human relationships with machines
- Danger of trusting AI with decisions it doesn't understand
- Questions about what makes us human

Discussion

Are these concerns still relevant today with ChatGPT, Claude, and other modern LLMs?
What has changed? What hasn't?

From ELIZA to Modern LLMs

ELIZA (1966):

- Pattern matching
- Hand-crafted rules
- No learning
- ~200 patterns
- Pure text manipulation

GPT-4 (2023):

- Neural networks
- Learned from data
- 1.7+ trillion parameters
- Statistical patterns
- Still text manipulation!

Think about it!

Both are fundamentally doing *pattern matching*—just at vastly different scales of complexity. Does the difference in scale create a difference in *kind*?

Why Start with ELIZA?

Pedagogical reasons:

- ① **Builds intuition:** Understand the basics before tackling complexity
- ② **Reveals assumptions:** See what "understanding" really requires
- ③ **Historical context:** Appreciate how far we've come
- ④ **Critical thinking:** Question what looks intelligent
- ⑤ **Hands-on learning:** You'll actually build it!

Core Insight

Even the most sophisticated language models are, at their core, finding and manipulating patterns. ELIZA makes this transparent.

Assignment 1: Build ELIZA

Your First Assignment

You will implement your own version of the ELIZA chatbot!

What you'll do:

- 1 Read rules from a configuration file
- 2 Implement pre-substitutions
- 3 Handle synonym substitutions
- 4 Implement pattern matching with regex
- 5 Implement decomposition and reassembly
- 6 Apply post-substitutions
- 7 Create a conversation loop

Assignment Details

Technical requirements:

- Python implementation
- Use provided `instructions.txt`
- Pattern matching with regex
- Proper text substitutions
- Conversation loop
- Greeting & goodbye messages

Deliverable:

- Single Jupyter notebook
- Runnable in Google Colab
- Markdown explanations
- Example conversations
- Well-documented code

Due Date

Check Canvas for the exact due date!

Tips for Success

- 1 **Start early!** Pattern matching can be tricky
- 2 **Test incrementally:** Build one piece at a time
- 3 **Use print statements:** Debug by seeing what patterns match
- 4 **Read Weizenbaum (1966):** Understand the original design
- 5 **Experiment:** Try having conversations with your bot
- 6 **Collaborate:** Discuss ideas with classmates (but write your own code!)
- 7 **Ask questions:** Use Discord and office hours

Think about it!

Pay attention to how it *feels* to talk to your ELIZA. Does it ever seem to "understand" you? When and why?

Learning Objectives

By completing this assignment, you will:

- Understand basic natural language processing
- Master regular expressions and pattern matching
- Appreciate the difference between simulation and understanding
- Gain hands-on experience with conversational AI
- Develop critical thinking about AI capabilities
- Build a foundation for more complex models

Remember

This is just the beginning! Each week builds on the previous, taking us from simple pattern matching to state-of-the-art transformer models.

Key Takeaways

- 1 Language and thought are **related but separable**
- 2 Pattern matching can create the **illusion of understanding**
- 3 The **ELIZA effect** shows our tendency to anthropomorphize
- 4 Modern LLMs are more sophisticated, but still fundamentally doing **pattern matching**
- 5 Building these systems yourself reveals their **true nature**
- 6 Critical thinking about AI is **essential**

The Big Question

If something acts intelligent, talks intelligently, and helps intelligently... does it matter whether it's "really" intelligent?

Next Week Preview

Week 2: Computational Linguistics

We'll explore:

- Data cleaning and web scraping
- Tokenization (how to split text into pieces)
- Part of speech tagging
- Sentiment analysis
- Statistical learning in language acquisition

Reading for next week:

- Sennrich et al. (2016) - Neural Machine Translation with BPE
- Kudo & Richardson (2018) - SentencePiece
- Saffran et al. (1996) - Statistical learning by infants

Readings for This Week

Required Readings

- 1 **Weizenbaum (1966)**: ELIZA — A Computer Program For the Study of Natural Language Communication Between Man and Machine [PDF]
- 2 **Fedorenko et al. (2024)**: The language network as a natural kind within the broader landscape of the human brain [Nature]
- 3 **Lupyan et al. (2020)**: Effects of language on visual perception [TICS]

Tip: Start with Weizenbaum—it will help you with Assignment 1!

Let's discuss!

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Office Hours: By appointment (Moore Hall 349)

Happy coding!